Fahim Arif

Backend & Full-Stack · Node.js · TypeScript · Python

Dhaka, Bangladesh · Remote · 4+ hrs overlap with US Eastern

fahim.cuet77@gmail.com · linkedin.com/in/fahim-arif · github.com/fahim-arif

Professional Summary

Senior software engineer with 7+ years building backend services and full-stack products in Node.js and TypeScript, including 3 years on a large video streaming platform. I design REST APIs, data models and service integrations on PostgreSQL, Redis and AWS. I debug production issues end to end — from API data defects to DRM, ad-insertion and adaptive-streaming failures. Fully remote for 7 years, including 3 years with a US-based team across time zones.

Technologies and Languages

- **Languages:** TypeScript, JavaScript, Python
- **Backend & APIs:** Node.js, NestJS, Express, Django, REST API design, background jobs (BullMQ)
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Prisma
- **Cloud & DevOps**: AWS (EC2, S3, Lambda, RDS, Cognito, CloudFront), Docker, Docker Compose, CI/CD (Azure DevOps)
- **Frontend:** React, Next.js, Redux
- **Security**: JWT, OAuth2, HttpOnly-cookie sessions, CORS, rate limiting, OWASP fundamentals
- **Testing & Tools:** Jest, Cypress, Git, Jira; Segment, Mixpanel, GA4
- **Streaming & Media Systems**: Shaka Player, hls.js, HLS/DASH, DRM (FairPlay, Widevine), AWS MediaPackage, AWS MediaTailor (SSAI), CloudFront CDN, EPG systems

Work Experience

Senior Software Engineer (Contract, Remote) **FreeCast** **Jul 2023 – Aug 2026**

- Optimized slow endpoints in FreeCast's Python backend (Django) through query optimization, database indexing, N+1 query fixes, Redis caching and pagination, cutting response times from a few seconds to a few hundred milliseconds.
- Traced a catalog API defect that served mismatched titles under a live network filter; isolated the root cause through systematic API testing and delivered the fix spec to the backend team.
- Designed the auth token hardening plan: moved access tokens to memory, added bearer-token origin allowlisting, and proposed backend-issued HttpOnly refresh-token cookies to close an XSS exposure

window.
- Replaced browser-local onboarding state with a server-side user flag, making intro-flow state consistent across browsers and devices.
- Reworked VOD package and entitlement resolution for the Curiosity Stream package launch, changing how subscription packages map to upgrade offers.
- Resolved two production playback failures at the protocol level: a DASH ad-insertion stall caused by a codec-string mismatch between AWS MediaPackage and MediaTailor manifests, and a Safari FairPlay DRM failure caused by EME/MediaKeys state corruption.
- Migrated playback from the legacy platform to the new streaming architecture.
- Built EPG systems handling live schedule data and real-time updates, and implemented Segment/Mixpanel/GA4 watch-time and engagement events, eliminating duplicate firing.

Senior Full Stack Developer (Contract, Remote) **CUDO B2Work** **Mar 2023 – Aug 2023**

- Built CI/CD pipelines on Azure DevOps with Docker and Kubernetes, automating builds and deployments.
- Built backend and UI for real-time collaboration and Gantt scheduling features.
- Built premium subscription features for the platform's paid tier.

Senior Full Stack Developer (Remote) **Civilized Software** **Feb 2022 – Mar 2023**

- Designed and built REST APIs and business logic on Node.js/Express, moving core platform logic out of the client application.
- Integrated video playback into the platform, increasing user engagement by 35%.
- Integrated Ethereum and Polygon smart contracts into an NFT marketplace.

Full Stack Web Developer (Contract, Remote) **Twomatches.xyz** **Oct 2021 – Feb 2022**

- Built and tested APIs in a microservice backend and deployed the platform on AWS.

Full Stack Developer (Remote) **Primates Development Pvt Ltd** **Jun 2019 – Sep 2021**

- Built automation tooling for Nokia manufacturing processes, improving production accuracy.
- Built an inventory management system that reduced material waste.

Projects

VOD streaming platform backend (in progress) https://github.com/fahim-arif/vod-platform-api

- Building the backend for a video-on-demand platform with NestJS, PostgreSQL (Prisma), Redis and BullMQ, run locally with Docker Compose.
- Modeled a hybrid catalog: browsable movies and series, seasons with their own metadata, episodes as children, and separate playable assets per title (trailers, alternate cuts).
- Designed availability windows by region, plan and licensing period; soft unpublish instead of hard delete to preserve watch history; internal integer keys with public IDs.

Education

- B.Sc. Computer Science, Chittagong University of Engineering & Technology **2016 – 2020**